

Event-State Theory: A Discrete Causal-Information Framework for Cosmology and Fundamental Physics

Torben Wille

Independent Researcher, Denmark

November 23, 2025

Abstract

Event-State Theory (EST) proposes that physical reality consists of a discrete sequence of static, fully realized informational configurations (“Event-States,” Ψ_n). Time is not a fundamental dimension but the emergent ordering of these frames. Transitions between frames are governed by a universal probabilistic optimization principle—the Path of Causal Least Resistance (PCLR)—implemented via a cost function J that jointly minimizes thermodynamic expenditure and loss of information fidelity.

Crucially, EST redefines the observable universe not as a singular explosion, but as an Asynchronous Multi-Point Crystallization (AMPC) event. The theory posits that the Big Bang occurred at multiple, temporally staggered nucleation sites. The framework naturally resolves the Hubble Tension via geometric interference, explains baryon asymmetry as the “scar tissue” of a primordial annihilation event, and offers a testable prediction of Pre-Causal Resonance—detectable vacuum fluctuations occurring prior to causal events ($t < 0$).

1 Introduction

Modern physics remains fractured between the smooth, deterministic geometry of General Relativity and the probabilistic, discrete character of Quantum Mechanics. Event-State Theory (EST) resolves this conflict by discarding the assumption of continuous time. “Motion” and “Time” are cinematic illusions produced by the rapid succession of static Event-States. The universe behaves as a computational engine with one overarching goal—the preservation of information—executed by selecting the successor state that achieves this goal at the lowest thermodynamic and informational cost.

2 Axioms of Discrete Reality

Axiom I — Discrete Causal Frames (Ψ_n)

Reality consists of a sequence of globally defined, static states

$$\Psi_n \in \mathcal{H}, \quad n \in \mathbb{N}, \tag{1}$$

where $\mathcal{H}$ is the configuration space (or generalized Hilbert space) of all admissible universal states. Each Ψ_n completely specifies all realized physical information (particles, fields, geometry) within a closed causal domain.

Axiom II — Field of Potential (Ω)

Between any two consecutive crystallized frames Ψ_n and Ψ_{n+1} , the system exists in a field of potential successor states

$$\Omega(\Psi_n) = \{\psi_1, \psi_2, \dots\} \quad (2)$$

containing every physically admissible next configuration consistent with conservation laws and causal structure. Change is not intra-frame motion but the discrete difference in informational topology between Ψ_n and Ψ_{n+1} .

3 Formal Framework: The Path of Causal Least Resistance

3.1 The Cost Function J

The resistance of a transition $\Psi_n \rightarrow \psi$ is quantified by the cost function

$$J(\Psi_n \rightarrow \psi) = \alpha \Delta E + \beta (1 - I_F) + \epsilon_{MM\bar{M}}(\psi), \quad (3)$$

where:

- ΔE is the Hamiltonian difference (thermodynamic cost) required to reconfigure the state.
- I_F is a Fisher-information-based functional measuring the preservation of particle distinguishability and informational structure between Ψ_n and ψ (information fidelity). Higher I_F corresponds to lower informational cost.
- $\epsilon_{MM\bar{M}}(\psi)$ is the *Primordial Asymmetry Constant* term. It represents the informational “scar tissue” surviving the pre-universal matter–antimatter annihilation and acts as a deterministic bias that tilts all probability distributions toward matter-dominant successor states. More precisely, $\epsilon_{MM\bar{M}}(\psi)$ takes slightly different values for matter-favoring and antimatter-favoring transitions such that, all else equal,

$$J(\Psi_n \rightarrow \psi_M) < J(\Psi_n \rightarrow \psi_{\bar{M}}), \quad (4)$$

ensuring a systematic, though small, preference for matter-dominated outcomes.

3.2 Probabilistic Selection Rule

The successor state is chosen according to a Boltzmann-like distribution over the field of potential:

$$P(\Psi_{n+1} = \psi) = \frac{1}{Z(\Psi_n)} \exp\left(-\frac{J(\Psi_n \rightarrow \psi)}{\lambda}\right), \quad (5)$$

where $\lambda > 0$ is the fundamental causal temperature that sets the sharpness of the selection and $Z(\Psi_n)$ is the normalization factor (partition function):

$$Z(\Psi_n) = \sum_{\psi' \in \Omega(\Psi_n)} \exp\left(-\frac{J(\Psi_n \rightarrow \psi')}{\lambda}\right). \quad (6)$$

For macroscopically large systems with many degrees of freedom and sufficiently small λ , the distribution in Eq. (5) is sharply peaked around the global minimum of J , leading to effectively deterministic evolution at large scales.

4 Cosmological Implications

4.1 Baryon Asymmetry: The War and the Scar

The pre-big-bang vacuum contained equal potential for matter (M) and antimatter ($\bar{M}$). Their annihilation was the energy source that funded the initial crystallization of the universe. In EST, the surviving matter is not a random violation of symmetry; it is the informational residue ϵ —the ash of that annihilation—stabilized by the pressure of its own birth.

This residue enters the dynamics through the asymmetry term $\epsilon_{MM\bar{M}}(\psi)$ in the cost function (3), which biases the Path of Causal Least Resistance toward matter-dominant outcomes at every subsequent frame-selection step. Ordinary matter is thus interpreted as the “scar tissue” of the primordial annihilation event.

4.2 Asynchronous Multi-Point Crystallization (AMPC)

EST rejects a singular, instantaneous “Big Bang” in favor of multiple, temporally staggered nucleation events $C_1, C_2, C_3, \dots$ occurring asynchronously across the primordial field. This is the Asynchronous Multi-Point Crystallization (AMPC) picture.

The Void as Epicenter. Cosmic voids are identified as the geometric centers (nucleation sites) of these crystallization events. Expansion waves sweep matter outward, forming the high-density filaments of the Cosmic Web at the interference boundaries where expanding “bubbles” collide.

The Hubble Tension Solution. We reside within the KBC Void, interpreted as a local nucleation site. Because matter (informational resistance) has been swept toward the periphery, the local frame-transition rate appears accelerated compared to the cosmic average. The observed Hubble parameter $H(x)$ at location x can be modeled as an interference sum of expansion contributions from all nucleation centers:

$$H(x) \approx \sum_i A_i \frac{\Phi(|x - C_i|)}{|x - C_i|}, \quad (7)$$

where A_i encodes the amplitude of each nucleation event and Φ is a radial profile function of the expansion wave. This position dependence naturally leads to an anisotropic effective Hubble “constant,” reconciling early- and late-universe measurements.

Polychronic Universe. Since nucleation was asynchronous, different regions of the universe have existed for different durations. The so-called “Axis of Evil” (large-scale CMB anisotropy) is interpreted here as the observable signature of temporal offsets between crystallization bubbles in an AMPC universe.

5 Applied Mechanics and Testability

5.1 Quantum State Imprinting (QSI): Regenerative Causal Geometry

By imposing an external informational geometry (for example, a tuned acoustic or electromagnetic “blueprint field”), a system can be forced into a high-resistance state that would normally be exponentially suppressed under the natural PCLR dynamics (e.g., instantaneous freezing or vitrification).

Thermodynamic Mechanism. Contrary to classical cooling, QSI does not require the step-wise removal of heat over extended time. Instead, the system creates a *causal short-circuit*: the latent heat energy released by the phase transition is instantaneously consumed to pay the informational cost ΔJ of the forced state change. In this sense, the system “metabolizes” its own thermal energy to buy its new structure, subject always to global energy conservation.

5.2 The Causal Perturbation Experiment: Detecting Pre-Causal Resonance

In standard quantum field theory, a sensor array surrounding a vacuum pair-creation event at $t = 0$ will detect no perturbation until $t > 0$, when causal signals have had time to propagate at or below the speed of light.

In EST, the crystallization of an Event-State is a selection process involving the entire field of potential $\Omega(\Psi_n)$. Consequently, EST predicts that sufficiently sensitive quantum sensors will detect *Pre-Causal Resonance* (PCR): a correlated agitation of the vacuum potential for $t < 0$, immediately prior to the event.

This is not retro-causality in the sense of signals traveling backward in time. Rather, it is the detection of the informational “pressure” or loading associated with an impending event before it crystallizes into physical reality in the next Event-State. Although the correlated disturbance appears at $t < 0$ in the detector frame, it cannot be modulated by the experimenter and therefore cannot be used to transmit usable information faster than light; relativistic causality remains intact. The presence or absence of such pre-event correlated activity is the unique falsifiability criterion of Event-State Theory.

6 Conclusion

Event-State Theory replaces continuous spacetime with a discrete, computed sequence of informational states. It unifies the arrow of time with the Second Law of Thermodynamics, identifying both as manifestations of the system’s drive to minimize calculation cost under the Path of Causal Least Resistance.

The theory is quantitatively suggestive and empirically vulnerable: it explains the Hubble Tension as a geometric interference pattern in an AMPC universe, redefines the “Big Bang” as a staggered phase transition, and proposes that the vacuum itself resonates with the “shadow” of future events before they occur. We are not observers floating in a river of time; we are the persistent informational structures that the Path of Causal Least Resistance has preserved.

Author’s Note on Methodology

The conceptual core, the Multi-Point/Asynchronous Multi-Point Crystallization model, the scar-tissue baryogenesis picture, and the Event-State framework are the original intellectual contributions of Torben Wille. Advanced symbolic computation tools and large language models served as cognitive prosthetics for translating these ideas into rigorous mathematical syntax and standard academic structure. The author affirms full responsibility for the scientific claims presented.